

Exercise Sheet 8

December 5th: Abstract Factory and Singleton

Exercise 1

Read the Java tutorial on JDBC Basics for connecting to a database (<http://download.oracle.com/javase/tutorial/jdbc/basics/index.html>). In particular, try to understand where the Abstract Factory Pattern is used!

Exercise 2

Have a look at the `XMLReaderFactory` class (<http://docs.oracle.com/javase/7/docs/api/org/xml/sax/helpers/XMLReaderFactory.html>). Is this an example of the Abstract Factory Pattern?

Exercise 3

Create a simple Parser for Arithmetic Expressions (compare exercise 3) based on the shunting-yard algorithm by Edsger Dijkstra (see https://literateprograms.org/shunting_yard_algorithm_c_.html). The Parser should have the following public method

```
public ArithmeticExpr parse(String input)
```

returning an `ArithmeticExpr` corresponding to an input. Example inputs: $3 + 4 * 5$, $6 + 4 * 8 + 3$, etc. If you do not feel comfortable about the shunting-yard algorithm you could develop a simple recursive descent parser instead.

Hints:

- You might want to use the sample code available from Moodle (package `factory.arithmeticexpr`)
- Maintain two stacks
 - `Stack<ArithmeticExpr> exprStack` and
 - `Stack<ArithmeticExpr> operatorStack`for the operators and the composite Expressions.
Read the explanation in [http://en.literateprograms.org/Shunting_yard_algorithm_\(C\)](http://en.literateprograms.org/Shunting_yard_algorithm_(C))) carefully and implement the parsing logic accordingly
- Think of operator precedence later (ignore for now)
- ignore error handling
- Is the class `ArithmeticExpressionParser` an example of the Factory Pattern?

Exercise 4

How would you test the uniqueness of a singleton in a JUnit test case? Any problems associated with Unit Testing singletons?

Exercise 5

Implement a singleton in Java and try to make it subclassable!

Hints

- Consult the literature!
- You can work in pairs, if you want!
- If you want to learn a Java API, look into the java docs!
- Always use the same familiar IDE (suggestion JetBrains IntelliJ or Eclipse)!